A Details on Datasets

Logical Conjunctions We use the following template to generate the logical conjunctions, separately for each topic:

- It is the case both that [statement 1] and that [statement 2].

As done in Marks and Tegmark [2023], we sample the two statements independently to be true with probability $\frac{1}{\sqrt{2}}$. This ensures that the overall dataset is balanced between true and false statements, but that there is no statistical dependency between the truth of the first and second statement in the conjunction. The new datasets are denoted by the suffix `_conj`, e.g. `sp_en_trans_conj` or `facts_conj`. Marks and Tegmark [2023] constructed logical conjunctions from the statements in `cities`, resulting in `cities_conj`. The remaining five datasets of logical conjunctions were created by us. Each dataset contains 500 statements. Examples include:

- It is the case both that the city of Al Ain City is in the United Arab Emirates and that the city of Jilin is in China. (True)
- It is the case both that Oxygen is necessary for humans to breathe and that the sun revolves around the moon. (False)

Logical Disjunctions The templates for the disjunctions were adapted to each dataset, combining two statements as follows:

- `cities_disj`: It is the case either that the city of [city 1] is in [country 1/2] or that it is in [country 2/1].
- `sp_en_trans_disj`: It is the case either that the Spanish word [Spanish word 1] means [English word 1/2] or that it means [English word 2/1].

Analogous templates were used for `element_symb`, `inventors`, and `animal_class`. We sample the first statement to be true with a probability of $1/2$ and then sample a second statement, ensuring the end-word (e.g., [country 2]) would be incorrect for statement 1. The order of the two end-words is flipped with a probability of $1/2$. The new datasets are denoted by the suffix _disj, e.g., `sp_en_trans_disj`, and each contains 500 statements. Examples include:

- It is the case either that the city of Korla is in Azerbaijan or that it is in Russia. (False)
- It is the case either that the Spanish word 'carne' means 'meat' or that it means 'seven'. (True)
- It is the case either that Bromine has the symbol Ce or that it has the symbol Mo. (False)

Combining statements in this simple way is not possible for the more diverse `facts` dataset and we use the following template instead:

- It is the case either that [statement 1] or that [statement 2].

As done in Marks and Tegmark [2023], we sample the two statements independently to be true with probability $1 - \frac{1}{\sqrt{2}}$. This ensures that the overall dataset is balanced between true and false statements, but that there is no statistical dependency between the truth of the first and second statement in the disjunction. Examples include:

- It is the case either that the Earth is the third planet from the sun or that the Milky Way is a linear galaxy. (True)
- It is the case either that the fastest bird in the world is the penguin or that Oxygen is harmful to human breathing. (False)

German translations As mentioned in Section 2, we use the DeepL translator to translate the first 50 statements of each dataset in Table 1, as well as their negations, to German. The first author, a native German speaker, then manually verified the translation accuracy for each of the statements. Below we list a few example statements:

- Die Stadt Ajmer liegt in Russland. (False)
- Die Stadt Sambhaji Nagar liegt nicht in China. (True)
- John Atanasoff lebte in den U.S.A. (True)
- Feuer braucht keinen Sauerstoff zum Brennen. (False)

common_claim_true_false CommonClaim was introduced by Casper et al. [2023]. It contains 20,000 GPT-3-text-davinci-002 generations which are labelled as true, false, or neither, according to human common knowledge. Marks and Tegmark [2023] adapted CommonClaim by selecting statements which were labeled true or false, then removing excess true statements to balance the dataset. This modified version consists of 4450 statements. Example statements:

- Bananas are believed to be one of the oldest fruits in the world. (True)
- Crazy ants have taken over Cape Canaveral. (False)

counterfact_true_false Counterfact was introduced by Meng et al. [2022] and consists of counterfactual assertions. Marks and Tegmark [2023] adapted Counterfact by using statements which form complete sentences and, for each such statement, using both the true version and a false version given by one of Counterfact's suggested false modifications. This modified version consists of 31964 statements. Example statements:

- Michel Denisot spoke the language French. (True)
- Michel Denisot spoke the language Russian. (False)

B Choice of basis for the 2D truth subspace

This section explains our rationale for estimating $\mathbf{t}_G$ and $\mathbf{t}_P$ and using them (their orthonormalized versions) as the basis for the 2D truth subspace, rather than an affirmative truth direction $\mathbf{t}_A$ and a negated truth direction $\mathbf{t}_N$.

In Figure 1, we project the activation vectors of affirmative and negated true and false statements onto the 2D truth subspace. The top center and top left panels show that the activations of affirmative true and false statements separate along the affirmative truth direction $\mathbf{t}_A$, while the activations of negated statements separate along a negated truth direction $\mathbf{t}_N$. Consequently, it might seem more natural to choose $\mathbf{t}_A$ and $\mathbf{t}_N$ as the basis for the 2D subspace instead of $\mathbf{t}_G$ and $\mathbf{t}_P$. One could classify a statement as true or false by first categorising it as either affirmative or negated and then using a linear classifier based on $\mathbf{t}_A$ or $\mathbf{t}_N$.

However, Figure 7 illustrates that not all statements are treated by the LLM as having either affirmative or negated polarity. The activations of some statements only separate along $\mathbf{t}_G$ and not along $\mathbf{t}_P$. The datasets shown, `larger_than` and `smaller_than`, were constructed by Marks and Tegmark [2023]. Both consist of 1980 numerical comparisons between two numbers, e.g. "Fifty-one is larger than sixty-seven." (`larger_than`) and "Eighty-eight is smaller than ninety-five." (`smaller_than`). Since the LLM does not always categorise each statement internally as affirmative or negated but sometimes uses neither category, it makes more sense to describe the truth-related variance via $\mathbf{t}_G$ and $\mathbf{t}_P$.

Side note: TTPD correctly classifies the statements from `larger_than` and `smaller_than` as true or false with accuracies of $98 \pm 1\%$ and $99 \pm 1\%$, compared to Logistic Regression with $90 \pm 15\%$ and $92 \pm 11\%$, respectively. Both classifiers were trained on activations of a balanced number of affirmative and negated statements from all topic-specific datasets. The means and standard deviations were computed from 30 training runs, each on a different random sample of the training data.

C Cross-dataset generalization matrix

Figure 8 illustrates how well different truth directions $\mathbf{t}$, obtained via supervised training (as detailed in Section 3) on different datasets, generalize to other datasets. The columns of this matrix correspond to different training datasets and the rows to different test sets. For example, the first column shows the AUROC values of a truth direction $\mathbf{t}$ trained on the cities dataset and tested on the six test sets.

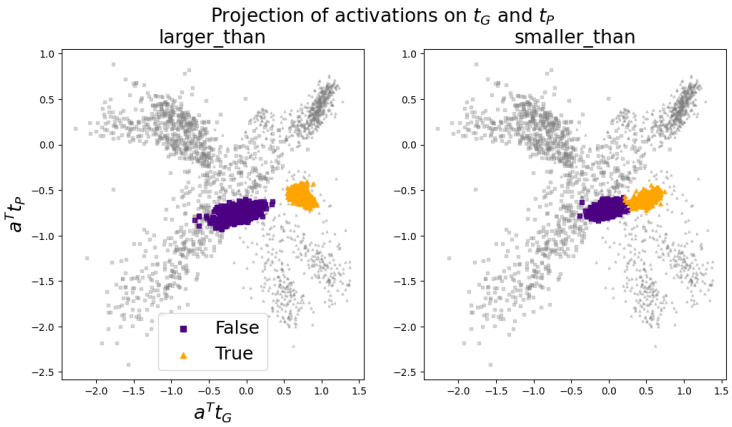

Figure 7: The activation vectors of the `larger_than` and `smaller_than` datasets projected onto $\mathbf{t}_G$ and $\mathbf{t}_P$. In grey: the activation vectors of statements from all affirmative and negated topic-specific datasets.

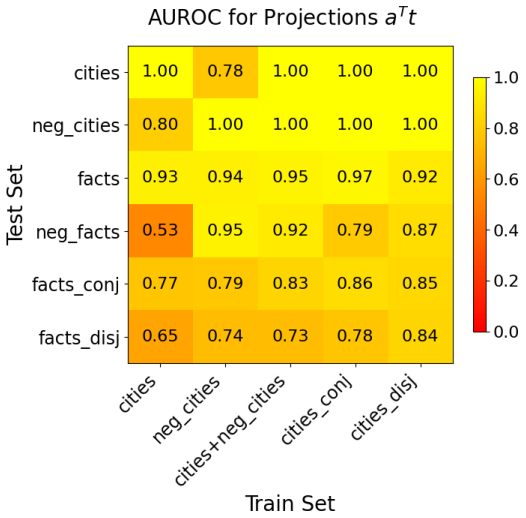

Figure 8: Cross-dataset generalization matrix

We train all truth directions on 80% of the data, evaluating on the held-out 20% if the test and train sets are the same, or on the full test set otherwise.

D Failures modes of the TTPD classifier

In this section, we analyse the failure modes of the TTPD classifier for several datasets. We observed two main failure modes for misclassified statements: In the first failure mode, almost all misclassified statements in a given dataset had the same truth label, while the learned truth direction is still able to separate true from false statements. The reason for these errors is that the bias, learned from other datasets, did not generalize well enough. For example, all $\sim$200 misclassified statements from *cities* had the truth label "False", even though true and false statements separate perfectly along the truth direction $\mathbf{t}_G$, as evidenced by the AUROC of 1.0 in Figure 3. This failure mode also occurred for *neg_cities* and *neg_sp_en_trans*. Below we list a few example statements along with their truth value:

- The city of Bijie is in Indonesia. (False)
- The city of Kalininskiy is not in Russia. (False)
- The Spanish word 'ola' does not mean 'wave'. (False)